

Bessel Functions of the First Kind

Power Series and Frobenius Method

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1 Motivation

Bessel functions arise when solving partial differential equations with radial symmetry, including Laplace's equation, the heat equation, and the wave equation in polar or cylindrical coordinates. They appear as solutions of *Bessel's equation*.

2 Bessel's Equation

For order $\nu \in \mathbb{R}$, Bessel's equation is

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0.$$

The point $x = 0$ is a *regular singular point*, making the Frobenius method appropriate.

3 Frobenius Series Ansatz

We seek a solution of the form

$$y(x) = \sum_{k=0}^{\infty} a_k x^{k+r}, \quad a_0 \neq 0.$$

Differentiating term-by-term gives

$$y'(x) = \sum_{k=0}^{\infty} a_k (k+r) x^{k+r-1}, \quad y''(x) = \sum_{k=0}^{\infty} a_k (k+r)(k+r-1) x^{k+r-2}.$$

4 Indicial Equation

Substituting into Bessel's equation and collecting the lowest power of x yields

$$a_0 [(r)(r-1) + r - \nu^2] = a_0 (r^2 - \nu^2).$$

Since $a_0 \neq 0$, we obtain the indicial equation

$$r^2 - \nu^2 = 0,$$

with roots $r = \pm \nu$.

5 Recurrence Relation

After collecting coefficients of x^{k+r} for $k \geq 0$, we find

$$a_{k+2}[(k+r+2)^2 - \nu^2] + a_k = 0.$$

Equivalently,

$$a_{k+2} = -\frac{1}{(k+r+2)^2 - \nu^2} a_k.$$

This two-step recurrence separates the solution into even and odd series.

6 Role of Initial Coefficients

Because the recurrence advances in steps of two:

- a_0 determines all even coefficients,
- a_1 determines all odd coefficients.

Choosing $r = \nu$ and setting $a_1 = 0$ yields the solution regular at the origin, defining J_ν .

7 Definition of J_ν

The Bessel function of the first kind is

$$J_\nu(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k + \nu + 1)} \left(\frac{x}{2}\right)^{2k+\nu}.$$

For integer $n \geq 0$,

$$J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! (n+k)!} \left(\frac{x}{2}\right)^{2k+n}.$$

8 The Second Solution: Y_ν

A second linearly independent solution is the Bessel function of the second kind, Y_ν . For noninteger ν ,

$$Y_\nu(x) = \frac{J_\nu(x) \cos(\nu\pi) - J_{-\nu}(x)}{\sin(\nu\pi)}.$$

As $x \rightarrow 0$,

$$Y_\nu(x) \sim x^{-\nu},$$

so Y_ν is singular at the origin. For integer order, logarithmic terms appear.

9 Summary

- Bessel's equation admits Frobenius series solutions
- Regularity at the origin selects J_ν
- Y_ν provides the second, singular solution
- These functions form the backbone of radial eigenfunction expansions